Jianhui (Duncan) Gong

Bioinformatics Ph.D. | The University of Michigan
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Ph.D. in Bioinformatics from the University of Michigan with research expertise in cancer proteogenomics, multi-omics integration, immuno-oncology, and translational bioinformatics. My current work focuses on integrating genomic, transcriptomic, proteomic, post-translational modification (PTM), immunopeptidomic, and clinical datasets to characterize tumor biology and identify biomarkers of treatment response. I have contributed to multiple large-scale cancer proteogenomics studies, including CPTAC projects, with a first-author paper in press at Cancer Discovery and peer-reviewed publications in Cancer Discovery, PNAS, Nature Communications, Cell Research, Nature, Science, and Genome Research.

Research Interests

Cancer proteogenomics; immuno-oncology; biomarker discovery; multi-omics integration; computational pathology; machine learning for translational cancer research.

Education

Doctor of Philosophy in Bioinformatics | The University of Michigan | USA | 2021.08 - 2026.08

Master of Science in Bioinformatics | The University of Michigan | USA | 2021.08 - 2023.04

Master of Science in Computer Science | Georgia Institute of Technology | USA | 2022.08 - 2025.12

Bachelor of Engineering in Bioengineering | South China University of Technology | China | 2009.09 - 2013.07

Research and Professional Appointments

Graduate Research Assistant | The Michigan Center for Translational Pathology | Michigan Medicine | The University of Michigan | USA | 2022.06 - Present

Conducted Ph.D. dissertation research in cancer proteogenomics, with a focus on integrative multi-omics analysis and translational biomarker discovery.

Led computational analyses of large-scale CPTAC melanoma datasets and contributed to a first-author paper in press at Cancer Discovery.

Developed components of the Turn-key Precision Oncology (TPO) platform to support scalable and clinically interpretable tumor profiling.

Associate Research Scientist | BGI Research | China | 2021.01 - 2021.07

Conducted machine learning research in precision health, including cfDNA-based classification of critically ill COVID-19 patients and single-cell RNA-seq-based gestational age prediction.

Applied feature selection, regression, and classification models to identify predictive molecular features from high-dimensional omics datasets.

Head of Technology Optimization Team | China National GeneBank | China | 2019.01 - 2020.12

Led a computational and technology optimization team supporting synthetic genomics, genome design, and multi-omics research programs.

Directed workflow optimization, software development, data management, and computational strategy development for large-scale collaborative projects.

Associate Research Scientist | BGI Research | China | 2018.03 - 2018.12

Assistant Research Scientist | BGI Research | China | 2015.09 - 2018.02

Bioinformatics Analyst | BGI Research | China | 2013.07 - 2015.08

Advanced from Bioinformatics Analyst to Associate Research Scientist through increasing responsibilities in bioinformatics analysis, project execution, and project leadership.

Conducted and later led computational analyses for synthetic genomics, CRISPR/Cas9 genome editing, and comparative genomics projects.

Contributed to major collaborative studies in the Synthetic Yeast Genome Project (Sc2.0), human embryo gene editing, and genome rearrangement analysis.

Publications

14 publications, including 2 in press, and 2 manuscripts under revision. First-author paper in press at Cancer Discovery. Google Scholar citations: 2,511.

Gong J, Petralia F, Ruiz Cuevas MV, Tignor N, Jain R, Gritsenko MA, Kitata RB, Ma W, Hsiao Y, Deng Y, et al. A comprehensive proteogenomic, immunopeptidomic, and metabolomic atlas decodes immunogenicity in cutaneous melanoma. Cancer Discovery. In press.

Chowdhury S, Abdollahi K, Petralia F, Kennedy JJ, Gong J, Ma W, Ivey RG, Pletcher RC, Krek A, Moshiri AS, et al. Identification of Proteomic Subtypes with Differential Responses to Immune Checkpoint Inhibition in Melanoma. Clinical Cancer Research. In press.

Eyunni S, Chu SC, Guan ML, Louw M, Young E, Carson SE, Gong J, Cieslik M, Chinnaiyan AM, Parolia A. STAG2 loss amplifies EWS-FLI1-driven microsatellite enhancer activity promoting Ewing sarcoma aggressiveness. Proceedings of the National Academy of Sciences of the United States of America. 2026 Apr 14;123(15):e2537425123. doi: 10.1073/pnas.2537425123. Epub 2026 Apr 8. PMID: 41950086; PMCID: PMC13079922.

Gumpert N, Sagie S, Arnedo-Pac C, Babu T, Weller C, Gonzalez-Perez A, Wang Y, Michel Todó L, Levy R, Chen X, Greenberg P, Dayan-Rubinov M, Yakubovich E, Wasserman-Bartov T, Zerbib M, Gong J, Rebernick RJ, Oliveira Tercero A, Agundez Muriel L, Benedek G, Kedmi M, Oren R, Ben-Dor S, Levin Y, Troyanskaya OG, Munzur AD, Wyatt AW, Cieslik MP, Quigley DA, Van Allen EM, Anandasabapathy N, Mateo J, Yang X, Martínez-Jiménez F, Lopez-Bigas N, Samuels Y. Recurrent Immunogenic Neoantigens and Their Cognate T-cell Receptors in Treatment-Resistant Metastatic Prostate Cancer. Cancer Discovery. 2026 Feb 6;16(2):250-269. doi: 10.1158/2159-8290.CD-24-1213. PMID: 41056506; PMCID: PMC12749549.

Shen Y, Gao F, Wang Y, Wang Y, Zheng J, Gong J, Zhang J, Luo Z, Schindler D, Deng Y, Ding W, Lin T, Swidah R, Zhao H, Jiang S, Zeng C, Chen S, Chen T, Wang Y, Luo Y, Mitchell L, Bader JS, Zhang G, Shen X, Wang J, Fu X, Dai J, Boeke JD, Yang H, Xu X, Cai Y. Dissecting aneuploidy phenotypes by constructing Sc2.0 chromosome VII and SCRaMbLEing synthetic disomic yeast. Cell Genomics. 2023 Nov 9;3(11):100364. doi: 10.1016/j.xgen.2023.100364.

Liang D, Mikhalchenko A, Ma H, Marti Gutierrez N, Chen T, Lee Y, Park SW, Tippner-Hedges R, Koski A, Darby H, Li Y, Van Dyken C, Zhao H, Wu K, Zhang J, Hou Z, So S, Han J, Park J, Kim CJ, Zong K, Gong J, Yuan Y, Gu Y, Shen Y, Olson SB, Yang H, Battaglia D, O'Leary T, Krieg SA, Lee DM, Wu DH, Duell PB, Kaul S, Kim JS, Heitner SB, Kang E, Chen ZJ, Amato P, Mitalipov S. Limitations of gene editing assessments in human preimplantation embryos. Nature Communications. 2023 Mar 7;14(1):1219. doi: 10.1038/s41467-023-36820-6.

Suzuki K, Yamamoto M, Hernandez-Benitez R, Li Z, Wei C, Soligalla RD, Aizawa E, Hatanaka F, Kurita M, Reddy P, Ocampo A, Hishida T, Sakurai M, Nemeth AN, Nuñez Delicado E, Campistol JM, Magistretti P, Guillen P, Rodriguez Esteban C, Gong J, Yuan Y, Gu Y, Liu GH, López-Otín C, Wu J, Zhang K, Izpisua Belmonte JC. Precise in vivo genome editing via single homology arm donor mediated intron-targeting gene integration for genetic disease correction. Cell Research. 2019 Oct;29(10):804-819. doi: 10.1038/s41422-019-0213-0.

Ma H, Marti-Gutierrez N, Park SW, Wu J, Hayama T, Darby H, Van Dyken C, Li Y, Koski A, Liang D, Suzuki K, Gu Y, Gong J, Xu X, Ahmed R, Lee Y, Kang E, Ji D, Park AR, Kim D, Kim ST, Heitner SB, Battaglia D, Krieg SA, Lee DM, Wu DH, Wolf DP, Amato P, Kaul S, Belmonte JCI, Kim JS, Mitalipov S. Ma et al. reply. Nature. 2018 Aug;560(7717):E10-E23. doi: 10.1038/s41586-018-0381-y

Ma H, Marti-Gutierrez N, Park SW, Wu J, Lee Y, Suzuki K, Koski A, Ji D, Hayama T, Ahmed R, Darby H, Van Dyken C, Li Y, Kang E, Park AR, Kim D, Kim ST, Gong J, Gu Y, Xu X, Battaglia D, Krieg SA, Lee DM, Wu DH, Wolf DP, Heitner SB, Belmonte JCI, Amato P, Kim JS, Kaul S, Mitalipov S. Correction of a pathogenic gene mutation in human embryos. Nature. 2017 Aug 24;548(7668):413-419. doi: 10.1038/nature23305.

Jiang S, Liu Y, Xu C, Wang Y, Gong J, Shen Y, Wu Q, Boeke JD, Dai J. Dissecting Nucleosome Function with a Comprehensive Histone H2A and H2B Mutant Library. G3 (Bethesda, Md.). 2017 Dec 4;7(12):3857-3866. doi: 10.1534/g3.117.300252.

Zhang W, Zhao G, Luo Z, Lin Y, Wang L, Guo Y, Wang A, Jiang S, Jiang Q, Gong J, Wang Y, Hou S, Huang J, Li T, Qin Y, Dong J, Qin Q, Zhang J, Zou X, He X, Zhao L, Xiao Y, Xu M, Cheng E, Huang N, Zhou T, Shen Y, Walker R, Luo Y, Kuang Z, Mitchell LA, Yang K, Richardson SM, Wu Y, Li BZ, Yuan YJ, Yang H, Lin J, Chen GQ, Wu Q, Bader JS, Cai Y, Boeke JD, Dai J. Engineering the ribosomal DNA in a megabase synthetic chromosome. Science. 2017 Mar 10;355(6329). doi: 10.1126/science.aaf3981.

Shen Y, Wang Y, Chen T, Gao F, Gong J, Abramczyk D, Walker R, Zhao H, Chen S, Liu W, Luo Y, Müller CA, Paul-Dubois-Taine A, Alver B, Stracquadanio G, Mitchell LA, Luo Z, Fan Y, Zhou B, Wen B, Tan F, Wang Y, Zi J, Xie Z, Li B, Yang K, Richardson SM, Jiang H, French CE, Nieduszynski CA, Koszul R, Marston AL, Yuan Y, Wang J, Bader JS, Dai J, Boeke JD, Xu X, Cai Y, Yang H. Deep functional analysis of synII, a 770-kilobase synthetic yeast chromosome. Science. 2017 Mar 10;355(6329). doi: 10.1126/science.aaf4791.

Shen Y, Stracquadanio G, Wang Y, Yang K, Mitchell LA, Xue Y, Cai Y, Chen T, Dymond JS, Kang K, Gong J, Zeng X, Zhang Y, Li Y, Feng Q, Xu X, Wang J, Wang J, Yang H, Boeke JD, Bader JS. SCRaMbLE generates designed combinatorial stochastic diversity in synthetic chromosomes. Genome Research. 2016 Jan;26(1):36-49. doi: 10.1101/gr.193433.115.

Yan D, Wang Y, Murakami T, Shen Y, Gong J, Jiang H, Smith DR, Pombert JF, Dai J, Wu Q. Auxenochlorella protothecoides and Prototheca wickerhamii plastid genome sequences give insight into the origins of non-photosynthetic algae. Scientific Reports. 2015 Sep 25;5:14465. doi: 10.1038/srep14465.

Manuscripts Under Revision

Wang C, Clark NM, Jaehnig EJ, et al.; Clinical Proteomic Tumor Analysis Consortium (CPTAC). Multi-omic interrogation of soft tissue sarcoma reveals therapeutically relevant molecular subtypes. Cancer Cell. Under revision.

Jaehnig E, Huynh AT, Kaparos E, et al.; Clinical Proteomic Tumor Analysis Consortium (CPTAC). Comprehensive proteogenomic characterization of DNA damage response (DDR) across human tumors spanning ten cancer types. Clinical Cancer Research. Under revision.

Research Experience

Cancer Proteogenomics Projects

Mentor: Marcin Cieslik, Ph.D., Assistant Professor, The University of Michigan

CPTAC Melanoma Proteogenomics

Led integrative multi-omics analyses of large-scale CPTAC melanoma datasets, including genomic, transcriptomic, proteomic, PTM, immunopeptidomic, and clinical data.

Characterized molecular subtypes, genotype-phenotype associations, immune features, and clinical outcome associations in melanoma.

Contributed to biomarker discovery efforts for immunotherapy response and tumor biology.

TPO (Turn-key Precision Oncology) Platform

Contributed to the development of an integrative precision oncology analysis platform for rapid assessment of somatic mutations, copy-number alterations, structural variants, gene expression, and gene fusions.

Designed and implemented components for data organization and output storage to support scalable, reproducible, and clinically interpretable tumor profiling.

Machine Learning and Precision Health Projects

Mentor: Yong Bai, Ph.D., Director of the Big Data Intelligent Algorithm Lab, BGI

Diagnosis of critically ill COVID-19 patients using cfDNA sequencing features

Developed cfDNA motif-based classification models to distinguish critically ill COVID-19 patients, achieving an AUC of 0.81 for the submodel and 0.93 for the integrated model.

Gestational age prediction using PBMC single-cell RNA-seq data

Built cell-type-aware regression models to predict gestational age from PBMC single-cell RNA-seq data, achieving Pearson correlation coefficients of 0.80-0.88 across models.

HPV Integration Status Prediction

Mentor: Maureen Sartor, Ph.D., Professor, The University of Michigan

HPV integration status prediction using HPV and host gene expression

Developed machine learning models to predict HPV integration status from HPV and host-cell RNA expression profiles.

Applied combinatorial feature selection strategies, including Boruta, mutual information, and recursive feature elimination, and compared logistic regression with elastic net, random forest, and support vector machine models.

Computational Platform and Synthetic Biology Software Projects

Designed and implemented RiSE, a web-based resource information database to support wet-lab resource tracking, experimental organization, and synthetic biology project coordination.

Built TMS, a task management system to support synthesis manufacturing workflows, project tracking, and team-level coordination.

Contributed to Genome De Novo (Genovo), a web-based computer-aided design platform for de novo genome design.

Synthetic Genomics Projects

Mentor: Yue Shen, Ph.D., Director of the Synthetic Biology Lab, BGI

The Synthetic Yeast Genome Project (Sc2.0)

Conducted bioinformatics and multi-omics analyses for multiple Synthetic Yeast Genome Project (Sc2.0) studies, including synthetic chromosomes II, XII, and XIII.

Performed sequence validation, transcriptomic profiling, and genomic, proteomic, metabolomic, and lipidomic analyses to evaluate synthetic genome design and function.

Supported analyses of genome design-associated fitness defects, expression imbalance, aneuploidy-related phenotypes, and genome restructuring strategies.

CRISPR/Cas9 Genome Editing Projects

Mentor: Ying Gu, Ph.D., Director of the Super Cell Research Institute, BGI

Human Embryo and In Vivo Genome Editing

Performed whole-exome sequencing analysis of edited embryonic stem cell lines and identified potential off-target sites and variants to support safety assessment of CRISPR/Cas9-mediated human embryo editing.

Analyzed gene-correction frequency, indel efficiency, and large-deletion events in progeria mouse models to support computational characterization of SATI-mediated in vivo genome editing.

Comparative and Functional Genomics Projects

Mentor: Yun Wang, Ph.D., University of Copenhagen

Genome Rearrangement, Organelle Genome, and Mutant Library Analyses

Developed computational approaches to identify and characterize genome rearrangements generated by SCRaMbLE.

Performed organelle genome assembly, gene annotation, phylogenetic analysis, and mutant library composition analysis across comparative and functional genomics projects.

Mentoring Experience

BGI Innovation Class Student Mentorship | BGI Research / China National GeneBank | Shenzhen, China

Mentored five BGI Innovation Class students during their 1.5-year research training at BGI, providing guidance in bioinformatics analysis, synthetic genomics, multi-omics workflows, computational project design, and scientific communication.

Trained students in research project execution, data analysis, literature interpretation, result presentation, and collaborative manuscript-oriented research.

Former mentees have continued to Ph.D. programs at institutions including the University of Chicago, Southeast University, Shanghai Jiao Tong University, and Tsinghua University.

Computational Skills

Programming: Python, R, Perl, C/C++.

Bioinformatics: WGS, WES, targeted sequencing, RNA-seq, miRNA-seq, proteomics, phosphoproteomics, ubiquitylomics, glycoproteomics, acetylomics, HLA immunopeptidomics, metabolomics, lipidomics, single-cell RNA-seq, somatic mutation analysis, variant calling, copy-number alteration analysis, structural variant analysis, gene fusion analysis, and integrative multi-omics analysis.

Machine Learning: elastic net, random forest, support vector machines (SVM), gradient boosting, feature selection, regression and classification modeling, and model evaluation.

Pipeline & Software Development: reproducible workflow development, scalable computational analysis pipelines, database-backed web applications, and omics data management platforms.

Software Copyrights

Gong J, Wang Y, Sun C, Hu Z: Syntrans: Synthetic genome transcriptomic analysis software. Issued on Jul 23, 2018 (2018SR575563).

Zhang J, Gong J: GDOP: gRNA design and off-target prediction software. Issued on Jul 23, 2018 (2018SR575560).

Wang Y, Gong J, Shen Y, Chen T, Gao F: Synval: Synthetic genome sequence validation software. Issued on Mar 23, 2015 (2015SR05118).

Gong J, Wang Y, Shen Y: SegMan: Chromosome segmentation design software. Issued on Feb 5, 2015 (2015SR026088).

Authorized Patents

Gong J, Wang Y, Shen Y, Deng Y: Method for structuring artificial chromosome in target cell and eukaryocyte cell. CN106282224B. Granted on May 1, 2020.

National Standard

Wang Y, Wang J, Shen Y, Fu B, Zhao H, Gong J, Chen T, Liu X, Du J, Xie Q, Niu C: General assessment of quality evaluation for synthesized genes. Issued on Aug 30, 2019 (GB/T 37873-2019).

Selected Presentations

AACR Annual Meeting | 2025

Semi-Annual Scientific Investigator Meeting | 2023, 2024, 2025

Genome Project-write/8th Annual Sc2.0 Meeting | 2019

Second International Summit on Human Genome Editing | 2018

Oregon Health & Science University Board of Directors Meeting | 2018

The Seventh International Meeting on Synthetic Biology | 2017

13th International Symposium on the Genetics of Industrial Microorganisms | 2016

Grant-Supported Research Projects

National High-tech R&D Program (863 Program) | 2012AA02A708 | Synthetic Yeast Genome | CNY 30.57 million. Role: project member and computational analysis contributor.

Guangdong Province Science and Technology Plan | 2017B090904014 | Guangdong Provincial Academician Workstation of BGI Synthetic Genomics | CNY 1 million. Role: project member and computational analysis contributor.

Guangdong Province Science and Technology Plan | 2017B030301011 | Guangdong Provincial Key Laboratory of Genome Read and Write. Role: project member and computational analysis contributor.

Research Impact and Public Recognition

Research contributions to human embryo gene editing and synthetic yeast genome projects were covered by major international media outlets, including The New York Times, The Washington Post, CNN, BBC, The Guardian, The Wall Street Journal, NPR, and Financial Times.

Contributed to research programs recognized among the Top 10 Scientific/Technological Advancements in China in 2017.

Honors and Awards

PIBS Fellowship | The University of Michigan | Ann Arbor, MI, USA

Tucker Fund | The University of Michigan | Ann Arbor, MI, USA

Bernard Maas Fellowship | The University of Michigan | Ann Arbor, MI, USA

Poster Award | Genome Project-write / 8th Annual Sc2.0 Meeting | New York University | New York, NY, USA

Annual Project Excellence Award | BGI | Shenzhen, China

Semi-annual Project Excellence Award | BGI | Shenzhen, China

iGEM Foundation Awards: Gold Medals, First Runner-up, and Best Wiki | Boston, MA, USA; Hong Kong

The TECHPOOL Scholarship | South China University of Technology | Guangzhou, China

Editorial and Peer Review Service

Guest Editor: PNAS Nexus.

Peer Reviewer: Briefings in Bioinformatics; eLife; JCO Precision Oncology; iScience; PLOS Genetics; Heliyon; PLOS ONE.